

No. of Functional Features Included in the Solution

Date	01-07-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Login	Allows users to securely register and log in to access the application.	Authentication Module	Done	High
2	Text Preprocessing	Cleans, tokenizes, and prepares user input text for emotion analysis.	NLP Module	Done	High
3	Emotion Detection	Detects learner emotions using BiLSTM and BERT deep learning models with confidence scores.	Emotion Detection Module	Done	High
4	Personalized Learning Support	Generates emotion-aware study guidance and motivational responses using Gemini AI.	AI Response Module	Done	High
5	Streamlit Dashboard	Displays user input, detected emotions, confidence scores, AI responses, and model comparison.	User Interface	Done	Medium
6	CSV Logging	Stores interaction history, emotion predictions, AI responses, and timestamps for future analysis.	Logging Module	Done	Medium
7	Analytics & Visualization	Provides emotion trends and confidence analysis through interactive Plotly charts.	Analytics Module	Done	Medium

8	Model Comparison	Enables comparison between BiLSTM and BERT prediction performance.	Model Evaluation Module	Done	Medium
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	User Authentication, Text Preprocessing, Emotion Detection, Gemini AI Integration, Streamlit Dashboard
Additional / Nice-to-Have Features	Analytics Dashboard, CSV Logging, Model Comparison
Features Tested & Verified	All 8 implemented features were tested and verified successfully

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Streamlit Dashboard, User Input Form
2	Backend / Logic	3	Text Preprocessing, Emotion Detection, AI Response Generation
3	Database / Storage	1	CSV Logging, Emotion Records Storage
4	API / Integration	1	Google Gemini AI Integration
5	Security / Authentication	1	User Registration & Login Authentication